U.S. Airline Flight Performance Dashboard (2015)

Executive Summary

This project analyzes over 1 million domestic U.S. flight records from 2015 to identify patterns in flight delays, cancellations, airline performance, and airport operations. Using SQL for data preparation and Power BI for visualization, an interactive dashboard was developed to provide actionable insights into airline and airport performance.

1. Introduction

Problem Statement

Flight delays and cancellations significantly impact passengers, airlines, airports, and the overall economy. Understanding the causes and trends behind these disruptions can help stakeholders improve operational efficiency and customer satisfaction.

Project Objectives

- Analyze flight delays and cancellations.
- Measure airline performance using key KPIs.
- Evaluate airport operational efficiency.
- Identify monthly trends in flight operations.
- Provide actionable recommendations.

2. Dataset Overview

Data Source

2015 U.S. Domestic Flights Dataset

Dataset Components

- Flights Data
- Airlines Data
- Airports Data

Total Records

- Total Flights: 1.05 Million+

Key Fields Used

- Airline
- Origin Airport
- Destination Airport
- Arrival Delay
- Departure Delay
- Cancelled
- Cancellation Reason
- Month
- State

3. Data Preparation

Data Cleaning

- Handled null values in delay columns.
- Converted data types appropriately.
- Replaced missing values where necessary.
- Standardized column formats.

Data Modeling

Relationships were established between flight, airline, and airport data to enable comprehensive analysis.

DAX Measures Created

1. Total Flights
2. Cancelled Flights
3. Average Arrival Delay
4. Average Departure Delay
5. Cancellation Rate

4. Dashboard Overview

The Power BI dashboard consists of:

KPI Cards

- Total Flights: 1.05M
- Cancelled Flights: 40.5K
- Average Arrival Delay: 7.6 Minutes
- Average Departure Delay: 11.3 Minutes
- Cancellation Rate: 3.86%

Visualizations

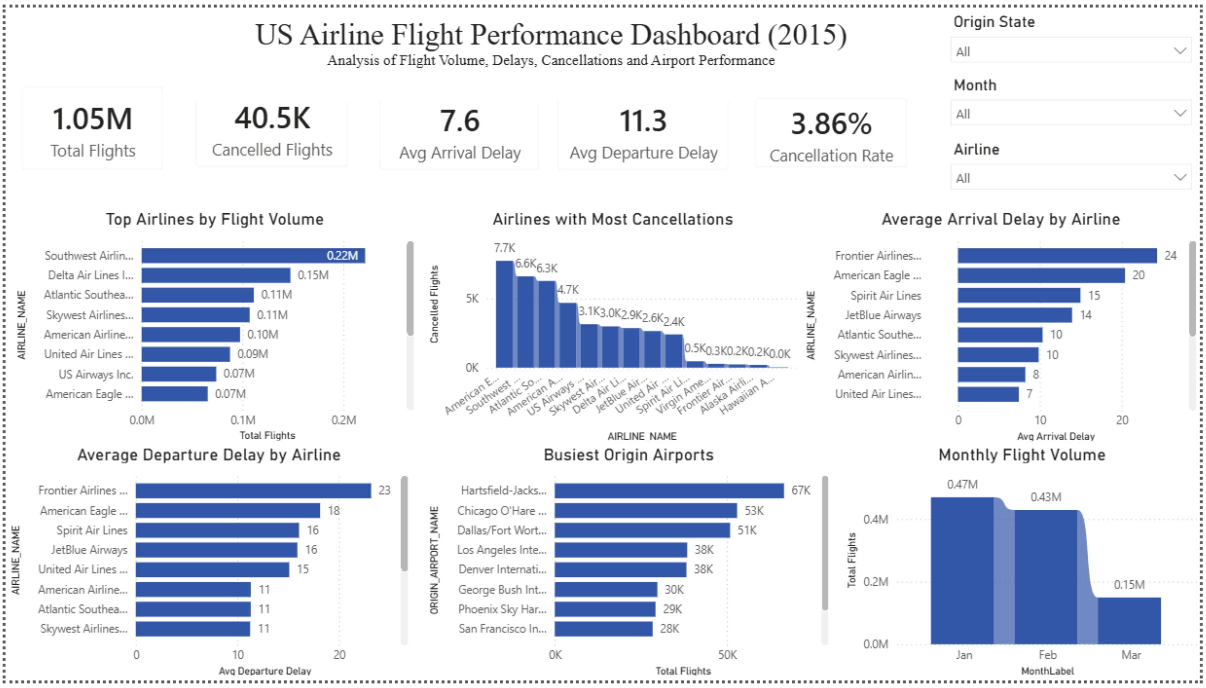
- Top Airlines by Flight Volume
- Airlines with Most Cancellations
- Average Arrival Delay by Airline
- Average Departure Delay by Airline
- Busiest Origin Airports
- Monthly Flight Volume

Filters

- Origin State
- Airline
- Month

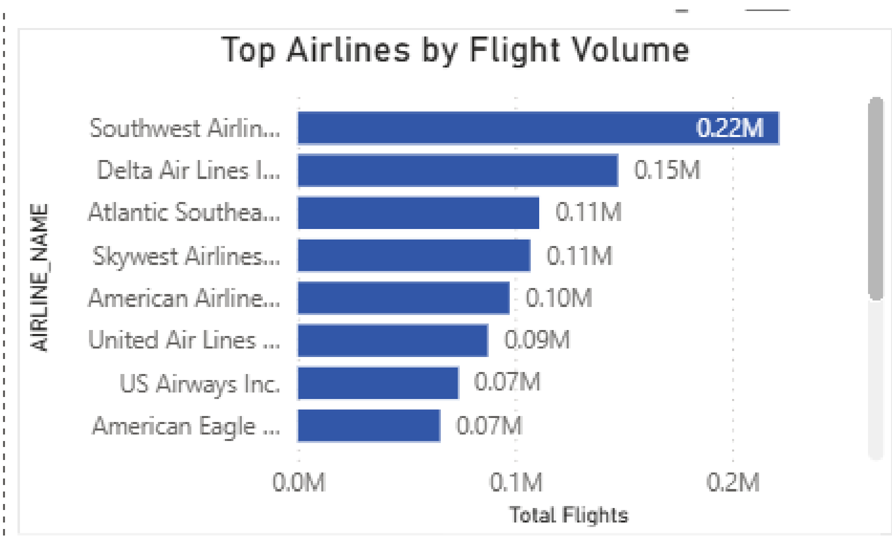
5. Key Findings

Finding 1: Complete Dashboard Overview



Analysis:
The dashboard provides a comprehensive overview of airline performance across multiple dimensions, including flight volume, cancellations, delays, airport performance, and monthly trends. Interactive filters allow users to analyze specific airlines, months, and states.

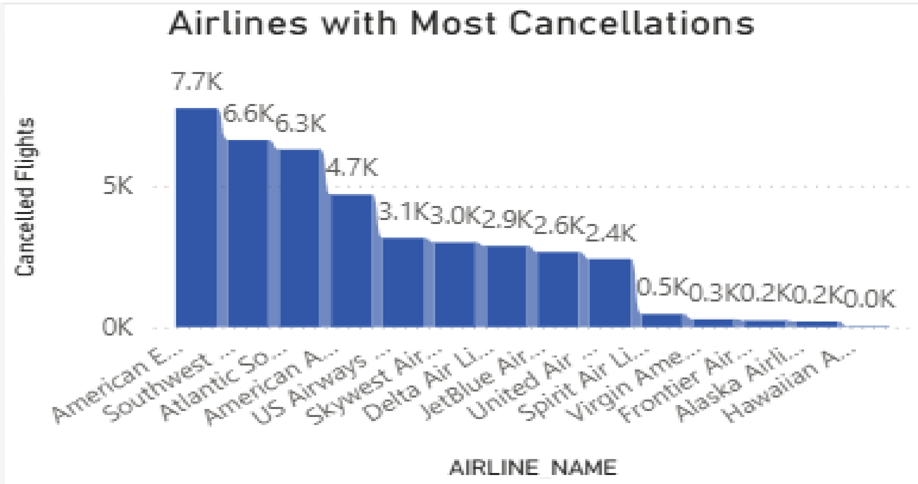
Finding 2: Top Airlines by Flight Volume



Analysis:
This chart shows the total number of flights operated by each airline. Southwest Airlines handled the highest number of flights, indicating a larger operational network and market presence compared to other airlines.

Business Insight:
High flight volume airlines require efficient scheduling and resource allocation to maintain operational performance.

Finding 3: Airlines with Most Cancellations

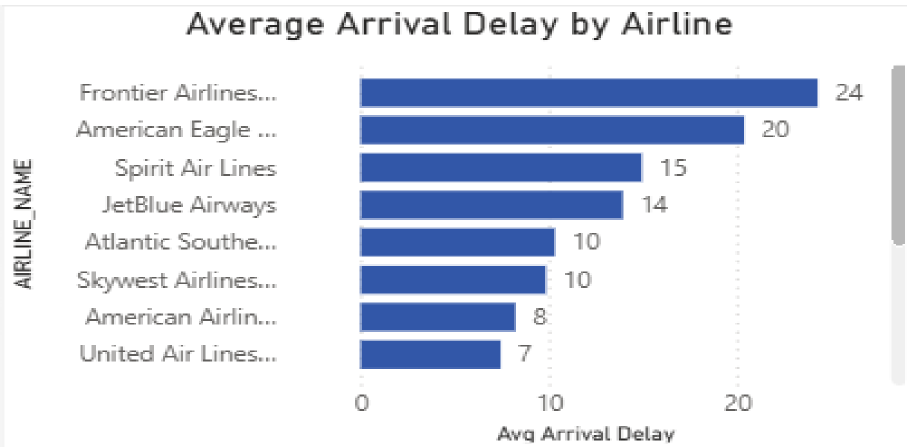


Analysis:
The chart highlights airlines with the highest number of flight cancellations. Certain airlines experienced significantly higher cancellation counts compared to competitors.

Business Insight:

High cancellation rates negatively impact customer satisfaction and operational efficiency. Airlines should investigate recurring causes of cancellations and implement preventive measures.

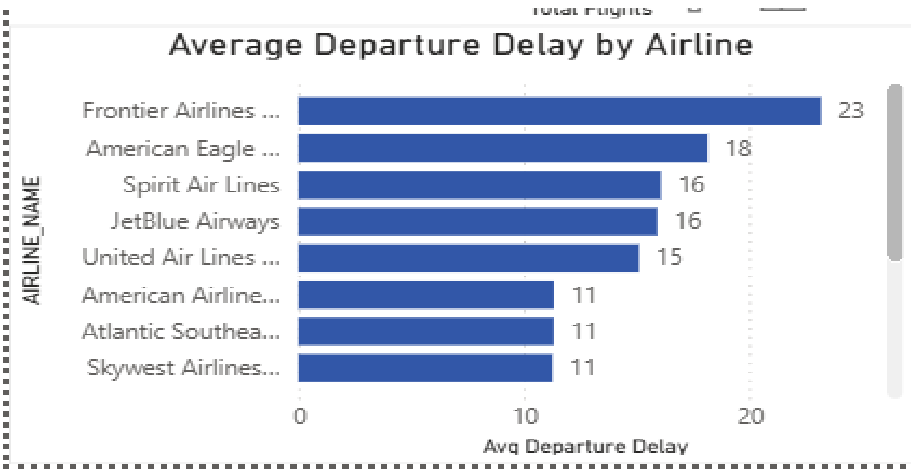
Finding 4: Average Arrival Delay by Airline



Analysis:
This visualization compares average arrival delays across airlines. Some airlines consistently experienced longer arrival delays, indicating operational inefficiencies or route-related challenges.

Business Insight:
Reducing arrival delays can improve customer satisfaction and enhance airline reputation.

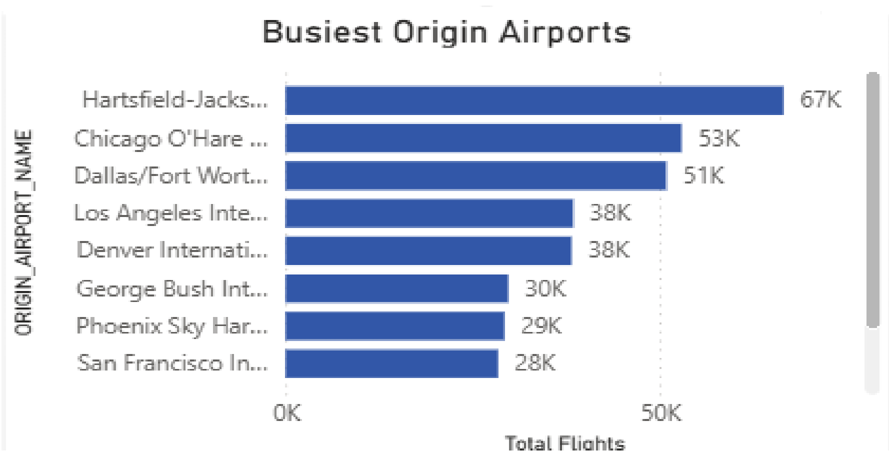
Finding 5: Average Departure Delay by Airline



Analysis:
The chart illustrates average departure delays for each airline. Departure delays often contribute directly to arrival delays and may indicate issues in boarding, maintenance, or airport operations.

Business Insight:
Improving turnaround times and operational planning can help reduce departure delays.

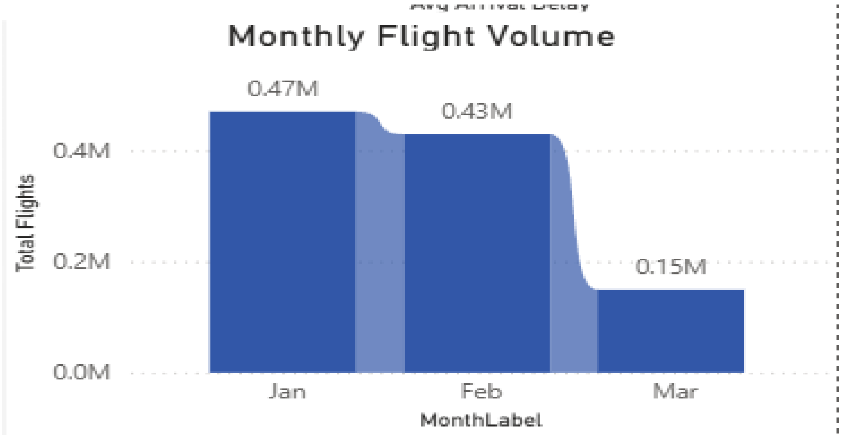
Figure 6: Busiest Origin Airports



Analysis:
This chart identifies airports with the highest number of departing flights. Airports such as Atlanta and Chicago handled significantly larger traffic volumes than others.

Business Insight:
High-traffic airports require optimized infrastructure and resource management to prevent congestion-related delays.

Figure 7: Monthly Flight Volume



Analysis:
The monthly trend chart illustrates changes in flight volume across the available months in the dataset. Flight activity varies over time due to seasonal demand and operational scheduling.

Business Insight:
Understanding monthly demand patterns helps airlines improve capacity planning and staffing decisions.

6. Recommendations

Recommendation 1

Airlines with consistently high departure delays should optimize boarding, ground handling, and aircraft turnaround processes.

Recommendation 2

Airport authorities should allocate additional resources to high-traffic airports to reduce congestion during peak periods.

Recommendation 3

Airlines should strengthen contingency planning during months with higher cancellation frequencies.

7. Conclusion

The analysis successfully identified major trends in airline performance, airport operations, delays, and cancellations. The dashboard enables stakeholders to monitor operational performance and make data-driven decisions to improve efficiency and passenger experience.

Tools Used

- SQL
- Power BI
- Microsoft Excel

Skills Demonstrated

- Data Cleaning
- Data Modeling
- DAX Calculations
- Dashboard Design
- KPI Development
- Data Visualization
- Business Analysis